

The Organization of Behavior

A Neuropsychological Theory

D.O. HEBB

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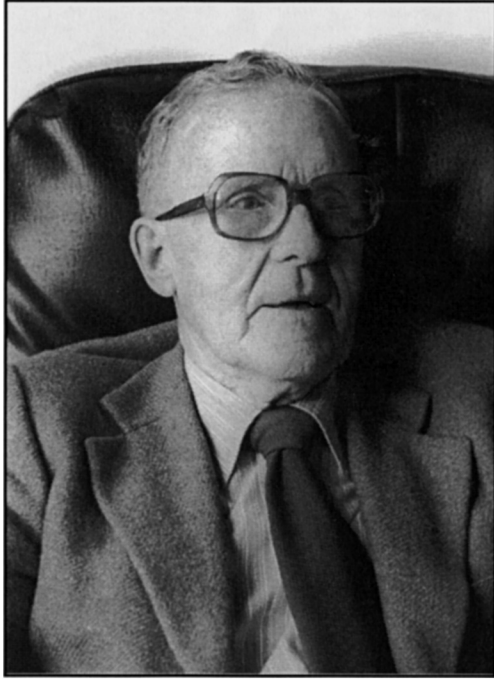
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*The Organization
of Behavior*

A NEUROPSYCHOLOGICAL THEORY

To
G.C.H., C.O.H.
and
A.M.H.



DONALD OLDING HEBB

Photograph by Richard E. Brown, April 1979.

Foreword

Richard E. Brown
Dalhousie University

Peter M. Milner
McGill University

Since the publication of *The Organization of Behavior* in 1949, it has become one of the most influential books in Psychology and Neuroscience (over 4100 SCI citations since 1989). According to Adams (1998, p. 419): “Two of the most influential books in the history of biology are Darwin’s *On the Origin of Species* (1859) and Hebb’s *Organization of Behavior* (1949). The books are famous because of the simple but powerful ideas that they contain.” While Darwin’s book can be purchased in any bookshop, Hebb’s book has been out of print since 1966, ensuring that Hebb’s comment that a classic normally means “cited but not read” (Fentress, 1987, p. 105) would be true in his case. One of us (REB) found this situation intolerable and decided to organize a reprinting of this book. This new edition of *The Organization of Behavior*, therefore, rectifies a long-standing problem for behavioral neuroscientists—the inability to obtain one of the most cited publications in the field. A complete list of Hebb’s publications from 1925 to 1987 is also included.

The Organization of Behavior played a significant part in the development of physiological psychology; the investigation of the neural foundations of behavior (Fentress, 1999; Milner 1999). It introduced the idea of the “Hebb synapse” and the “Hebbian cell assembly.” The more frequently cited of these two ideas is the Hebb

synapse, but, as pointed out by Seung (2000, p. 1166): “Hebb’s synapse is not his only legacy. Even after half a century, his theory as a whole is still inspiring, because it is a general framework for relating behavior to synaptic organization through the dynamics of neural networks.” Hebb was amused that as a result of his neurophysiological postulate, learning synapses had acquired his name—“because this postulate is one of the few aspects of the theory he did not consider completely original. Something like it had been proposed by many psychologists before him, including Freud in his early years as a neurobiologist.” (Milner, 1993, p. 127).

1. A Short Biography of Donald O. Hebb

Donald Olding Hebb was born on July 22nd, 1904 in the village of Chester, Nova Scotia, Canada. Both of his parents were physicians and his mother was the third woman to obtain an M.D. from the medical school of Dalhousie University in Halifax, Nova Scotia. His earliest ambition was to write a successful novel so he studied English and Philosophy at Dalhousie University, graduating in 1925. Although his degree was in Arts, he also took classes in chemistry, physics and mathematics, obtaining a second-class distinction in mathematics and a first class distinction in physics. He spent the next year teaching at the high-school he had attended in Chester. During this time he discovered Freud and James and began to study psychology. He then moved to Montreal and in 1929 became a part-time graduate student at McGill University, taking a graduate seminar in systematic psychology (Dalhousie University had no Department of Psychology at the time). To support himself he took a job as a teacher and after a year he was promoted to principal of an elementary school in Montreal. During the next few years he became an enthusiastic reformer of teaching methods [1; numbers refer to Hebb’s publication list], but his reforms were not always appreciated and he became a full-time graduate student in psychology at McGill in 1931. He completed an MA thesis, entitled “Conditioned and Unconditioned Reflexes and Inhibition” in April 1932. Although Hebb [112, p. 283], in the light of subsequent work, dismissed this MA thesis as nonsense, it does contain original ideas and was passed “cum

laude” by two examiners. As the thesis was theoretical, Hebb was encouraged to do experimental work, which he started under the supervision of one of his examiners, Dr. Boris Babkin, a former student of Pavlov.

During the writing of his MA thesis, Hebb was bedridden with a tubercular hip, and his experimental work ended when his wife was killed in a car accident. He became depressed and disenchanted with the direction of his research and the McGill psychology department. Encouraged by Professor Babkin, Hebb wrote to Karl Lashley at the University of Chicago and was accepted there as a PhD student in July 1934. Lashley was an eminent and respected physiological psychologist. In fact, during the 1920’s his laboratory was almost the only one in America where genuine research on the neural bases of behavior was being done. He was at the peak of his career when Hebb became his student. Hebb learned a great deal during the year he spent in Chicago, with lectures from such giants as Thurstone, Köhler, Herrick, and Lashley.

The next year Lashley was offered a professorship at Harvard and Hebb obtained permission to accompany him, so his PhD was awarded by Harvard University. His PhD thesis (Hebb, 1936) examined the innate organization of visual perception in the rat and the results of this research were published in a series of three papers [2, 3, 4]. In these papers, Hebb concluded that some aspects of rat vision are innate, though by the time he wrote this book he had reverted to a more empiricist position. Hebb remained at Harvard for another year as a teaching assistant and to complete research on place learning in rats that he had started in Chicago.

His first appointment after leaving Harvard (1937) was a two-year fellowship with the neurosurgeon Wilder Penfield, director of the newly established Montreal Neurological Institute. His responsibility was to investigate intellectual deficits in patients with cortical lesions, mostly those resulting from operations performed for the relief of epilepsy. He was particularly intrigued by the finding that patients with large frontal-lobe lesions had normal intelligence-test scores, some even improved after operation. He also detected visual impairment in a patient after right temporal lobectomy, a cortical region not previously associated with vision.

Hebb's work with Penfield at the Montreal Neurological Institute developed his interest in human brain function and the behavioral effects of neurological damage [9], especially with respect to intelligence [7, 8, 13] and initiated a long-term collaboration between the two men. Indeed, Hebb's student, Brenda Milner, began her work with Penfield and went on to study H.M., surely one of the most famous patients in the history of neuropsychology.

At the end of his Fellowship Hebb accepted a position at Queen's University in Kingston, Ontario and studied the effect of cortical lesions on rat learning. He wrote a paper in which he made the unheard of proposal that early experience could permanently influence intelligence [18], a theme that runs through much of Hebb's later writing, including this book. He also developed the Hebb-Williams maze during this time, although the paper was not published until much later [34]. This work arose from Hebb's interest in intelligence testing and he was attempting to develop a Stanford-Binet test for rats.

In the meantime, Lashley had been appointed Director of the Yerkes Laboratories of Primate Biology in Orange Park, Florida. In 1942 he persuaded Hebb to join him in an investigation of the effects of brain lesions on intelligence and emotion in chimpanzees. During the five years he spent in Florida, Hebb devised tests of chimpanzee emotion as well as making some pioneer observations of dolphin behavior [38]. His work with chimpanzees developed his interest in emotionality [28] and led him to compare emotional problems in chimpanzees and humans [35]. Concerning his time working with chimpanzees, Hebb said that he "learned more about human beings during that time than in any other five-year period of my life, except the first." [112, p. 293]. It was during this period with Lashley and his colleagues that Hebb wrote most of *The Organization of Behavior*.

In 1947 Hebb returned to McGill [at a salary of \$5500 per year, plus \$500 moving expenses] as one of four new professors recruited to revive a psychology department that had all but ceased to exist during the war. The Chairman, Robert MacLeod, moved to Cornell the next year and Hebb occupied the Chair for the next ten years. When *The Organization of Behavior* was published in 1949, it was, to Hebb's astonishment, an immediate success, but

he was in no position to sit back and enjoy the acclaim. After the rather sedate pace of life in Orange Park he was now teaching three undergraduate courses, including the large introductory course, plus an evening graduate seminar, setting up animal labs and arranging funding for research on the effects of rearing on perception and intelligence in rats and dogs. He was also preparing an introductory textbook for his course [61]. This left him with little time, or incentive, to tinker with his neural theory or to revise it in the light of the revolutionary developments in neuroanatomy, neurophysiology and information theory that were taking place.

Under Hebb, McGill was at the forefront of the physiological approach to psychology. McGill graduates of that era were in demand in all parts of the world. Hebb's textbook, published in 1958, was distinctly physiological in scope and further contributed to the dissemination of his neural theories of mental processes. For the next 25 years he taught his notoriously rigorous graduate seminar (mandatory for every graduate student) and Introductory Psychology to classes of more than a thousand students. In 1970 he was appointed to a four-year term as Chancellor of McGill. He was successively elected President of the Canadian Psychological Association and the American Psychological Associations; he was also elected a Fellow of both the Royal Society of London and the American Academy of Arts and Sciences. He received many other honors, including innumerable honorary degrees. In 1976 he retired to his family home in Nova Scotia and was appointed honorary Professor at Dalhousie University in 1977. He died in 1985 following a hip replacement operation.

2. The Organization of Behavior

When did Hebb develop the ideas that went into *The Organization of Behavior*? The idea for the Hebb synapse first appeared in 1932 in his MA thesis, where he discussed the functioning of the synapse in conditioned reflexes. Hebb opens his introduction by stating that: "The purpose of this paper is to present a theory of the functioning of the synapse based on the experimental work of Sherrington and Pavlov, on reflexes and inhibitions. The implications of these things for psychological theory, in some aspects, has been far from clear. In

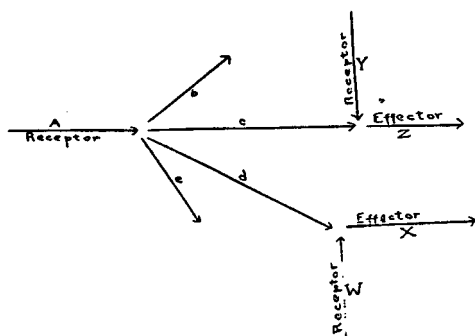


FIG. I

W-X } reflex arcs.
Y-Z }

A is another receptor.

b,c,d,e, connecting neurons.

(Arrowhead gives direction in which impulses
are transmitted.)

FIGURE 1.

looking for a firm basis for psychology in physiology, there are some peculiarities about the results of both investigators which demand serious consideration and suggest another interpretation of their work" (Hebb, 1932, p. 2).

In Figure 1 of his thesis, Hebb makes his first attempt to illustrate the synaptic changes associated with conditioning. The figure shows branches of a stimulus input going to two unconditioned reflex arcs. Pavlov's experiments show that if one of the reflex arcs is active when a stimulus is applied to the input, the branch going to that reflex becomes more potent; branches going to inactive effectors do not. Hebb went further, assuming that the potency of other branches declines. He summarized the first section of his thesis as follows: "An excited neuron tends to decrease its discharge to

inactive neurons, and increase this discharge to any active neuron, and therefore to form a route to it, whether there are intervening neurons between the two or not. With repetition this tendency is prepotent in the formation of neural routes" (Hebb, 1932, p. 13).

This may be seen as the emergence of the Hebb synapse idea. Hebb underlined it (the typewriter code for italics) as he later italicized the corresponding postulate in his book.

On November 24th 1934, Hebb submitted a revised version of his MA thesis as a term paper for Anatomy 316 (Elementary Neurology), which he took as a PhD student at the University of Chicago. This paper, entitled "The interpretation of experimental data on neural action" was noted as "A very thoughtful paper" by C.J.Herrick. In this paper, Hebb shows two diagrams of possible synaptic activity during conditioning and states that: "There is as yet no understanding theoretically of the process of conditioning, and when this is understood, it may throw light on the reflex activity of the [spinal] cord, and perhaps account for the existence of some of the apparently rigidly inherited patterns. The conception of block and facilitation at the synapse postulates a process which can strengthen and perpetuate a route once formed, but none whatever to account for the establishment of the route in the first place—a most important weakness in the whole theory" (Hebb, 1934, p. 15).

Hebb's first use of the term "Organization of Behavior" was in the title of two papers published in 1938 [5, 6] on the field orientation of the rat. In these papers Hebb examined the effects of cortical lesions on learning visual discriminations. Although these papers have the mark of Lashley on them, Hebb's focus was to develop a better behavioral test for the study of the effects of cerebral destruction. He states that: "It is clear that the success of the physiological analysis depends on the adequacy of the behavioral tests used. Much of the difficulty found in the evaluation of the effects of cerebral destruction, both clinically and with animal experimentation, is due to the fundamental difficulty of the analysis of behavior and to the unsatisfactory available accounts of it." [5, p. 333]. In light of recent research on context conditioning, it is interesting that these studies discussed the importance of room cues on the discrimination of objects in an open field.

During the years between 1934 and 1946, Hebb continued to work on his ideas about the organization of behavior. While still in Chicago in 1934, he wrote five chapters (90 pages) of a first draft of a book entitled "Scientific methods in psychology: A theory of epistemology based on objective methods in psychology." Parts of this book were to be included as a final chapter in *The Organization of Behavior* but were deleted during the editorial process.

In February, 1944 Hebb discovered that Lorente de Nó had recently shown that closed circuits were to be found throughout the brain, and "that one neuron by itself may not be able to excite a second neuron at the synapse, but can do so if supported by simultaneous action from another neuron" [112, p. 295]. It was these ideas (of neural loops) of Lorente de Nó that Hebb realized "were just what he needed to develop a more realistic theory of the mind" (Milner, 1993, p. 127). On 28 April, 1944, Hebb wrote to Lorente de Nó to ask if he could spend a month at the Rockefeller Institute to "become familiar with your work." When Lorente de Nó replied on 1 May 1944 that "At present my work is concerned with the relationship between the production of the nerve impulse and the metabolism of the nerve, a problem that is of little immediate interest to a psychologist," Hebb decided to delay his study trip to Rockefeller and wrote to Lorente de Nó (4 May 1944) that "I am profoundly convinced that psychological progress will depend on research such as yours...that psychological theory can go no farther without a more detailed basis in neurophysiological fact."

Hebb first outlined his ideas for *The Organization of Behavior* in five pages of typed notes in June 1944. These notes contain his early thoughts on the term "organization." In this definition, he considered both behavioral and physiological organization and considered the "concept of *set*," which he used to denote the pre-existing state of the nervous system encountered by an incoming stimulus. The second draft of *The Organization of Behavior* was developed in March–July 1945 and written under the title "Precis: The structure of a set of neuropsychological speculations." A complete draft of *The Organization of Behavior* was finished in 1946 and sent to Lashley, who wrote two pages of comments on the manuscript in February 1947. In June 1947, the book was submitted to Charles Thomas publishers. Henry Nissen received

the manuscript from Thomas on 21 June 1947. It was 150 pages long and insured for \$300.00.

Nissen edited this manuscript from June 1947 to September 1948, sending Hebb numerous comments and suggestions for changes. On 22 September, 1948, Hebb sent the final draft of the manuscript to the publishers. However, in January, 1949, Thomas decided not to publish the book, citing “circumstances beyond their control” and returned it to Hebb. After consulting with Beach, who suggested Hoeber (the publishers of his 1948 book, *Hormones and Behavior*), and considering Appleton-Century as publishers, Hebb sent the book to John Wiley and Sons, who accepted it with great praise on 18 March 1949 and published it in the fall of 1949.

On April 26, 1949, Hebb wrote to Lorente de Nó that “I believe that my book will be able to show that modern ideas in neurophysiology, and particularly some of those that you have developed, have a revolutionary significance for psychological theory.”

3. The Cell Assembly: Conception.

Ever since his observations of patients at the Montreal Neurological Institute, Hebb had puzzled over the nature of thought and the way ideas might be represented in the brain. The relevance to his problem of Lorente de Nó’s recurrent loops struck home to him in 1944, as mentioned earlier, and provided the key to his theory. Lashley (1942), in a paper written before Hebb arrived in Orange Park, raised the possibility that reverberations of Lorente de Nó’s loops might play a role in vision. His use of the loops as oscillators for his wave-interference theory was more limited and very different from Hebb’s grand scheme to represent concepts with them, but Hebb never refers to it in his book, which may be one reason for Lashley’s less than enthusiastic reception of it.

In the absence of evidence to the contrary, Hebb assumed that before they became cell assemblies, Lorente de Nó’s loops were randomly connected to sensory input. When a stimulus was impressed on the system some loops would receive more input than others, depending on the pattern of the connections, and they would therefore reverberate more intensely and persistently. After a

number of such episodes the connections within the most vigorously reverberating group of cells would be strengthened to such an extent that partial input, or even remote associations, would subsequently activate the whole assembly.

Hebb postulated that when an assembly was firing as a result of sensory input, its activity represented the perception of the stimulus; when it fired in the absence of the corresponding sensory input the activity represented the concept of the stimulus. When a sequence of stimuli occurred frequently, Hebb assumed that the aroused assemblies would become associated with each other via strengthened cross-linkages, to establish what he called *phase sequences*.

Having established the physical reality of ideas and their mode of association, Hebb was able to discuss expectancies, attention, attitudes, intelligence, even mental illness, in the rest of the book with a freedom not permitted to a serious psychologist in America for almost half a century. It is probable that this contributed to the popularity of the book among psychologists who had been deprived of such rich fare for too long. The earlier, more speculative chapters, on the other hand, attracted the attention of computer scientists, engineers, philosophers and biologists of all persuasions. Before its publication, Hebb used a manuscript of the book in his graduate seminar at McGill and it fell into the hands of one of us (PMM, who had spent the war working on radar and atomic energy). After reading a few chapters he applied to do graduate work with Hebb and was accepted.

Hebb's neurophysiological postulate (p. 62) was of particular interest to physiologists. Psychologists have always assumed that association requires contiguity of two or more events, but many physiologists thought that learning occurred when a connection was used repeatedly. As Hebb pointed out in his MA thesis, if such a synapse is initially ineffective it can never increase in strength, however often it fires. The implication of Hebb's postulate is that synapses increase in strength only if the post-synaptic neuron is fired by some other input at about the same time as a pre-synaptic impulse occurs. When synapses that behaved in this way were found, neurophysiologists called them "Hebb synapses" and the name has stuck.

Learning synapses in computer models of the brain are also often

referred to as Hebb synapses, though usually they do not adhere strictly to Hebb's definition. In order to prevent all synapses from increasing indefinitely in strength, it is additionally postulated that synaptic effectiveness is reduced when pre- and post-synaptic elements fire at separate times. It is interesting that in his 1932 MA thesis, the synaptic learning process described by Hebb (see page *xi* above) included a process by which synaptic strength was reduced when the post-synaptic membrane did not fire at the same time as the incoming impulse.

4. The Cell Assembly: Half a century later.

Hebb's notes indicate that the chapters in which the cell assembly and phase sequence are developed cost Hebb much time and effort, and he was aware that too often speculation had to be substituted for vital neurological and anatomical information that was not available. On page *xi* of the Introduction he gives the reader permission to skip those chapters. He argues strongly in defense of his neural speculation, but on page 80 he tells us that he expects the model to be amended in the future, as better information about the nervous system becomes available. The future is now with us, and it is possible to see that Hebb was not fortunate in his timing relative to the state of neurophysiological knowledge.

As the book was being published, information was coming to light concerning the importance of the reticular system of the brainstem for arousal. During the next few years the relevance to behavior of subcortical structures such as the amygdala, septal area and hippocampus began to be evident. Furthermore, as he worked on the book, Hebb accepted the widespread belief of most physiologists and avoided incorporating synaptic inhibition (though a theory to account for it was advanced in his MA thesis), only to have J.C.Eccles (Eccles et al., 1954), previously the most vociferous opponent of the idea, conclusively demonstrate its existence a year or two later.

Somewhat more recently, new genetic, histological and brain imaging techniques have fueled the rise of a modern phrenology, quite at variance with the anti-localization sentiment prevailing when this book was being written. Even if the present trend toward

nativism is sometimes exaggerated there seems little doubt that Hebb's assumption that cortical connections are initially random was overcautious. His argument that vision developed very slowly in human babies was based on two very dubious observations. The impaired vision of chimpanzees reared in the dark turned out to be caused by irreversible damage to the retina, and von Senden's reports of the difficulties of people given sight as adults after being born blind could be plausibly explained as damage to innate connections due to lack of normal stimulation during development.

Even the simple purity of Hebb's neurophysiological learning postulate has proved illusory. We now know that synaptic activity gives rise to a complex sequence of molecular changes, some brief, others lasting for hours, days, and sometimes permanently, that change the characteristics of both sides of the synaptic junction (e.g. Greengard, 2001). The inelegant reverberating loops of the cell assembly, almost as impervious to research as the mental activity they were meant to replace, are no longer needed to explain the persistence of thoughts. The stimulated neurons may continue to fire because of a persisting intrinsic change rather than by recurrent restimulation.

Without Hebb's speculation many of these discoveries might not have been made, or only after a longer delay. The biological speculations in chapters 4 and 5 of this book are interesting from a historical point of view, though as neuroscience they have now been largely superceded. Many of the later chapters, being based mainly on more reliable behavioral observations, continue to be valid and to the new reader they frequently bring the tingle of encountering a brilliant, and often witty, insight into a problem.

When he embarked upon the book, Hebb's original purpose was to explain in neural terms the processes whereby early experiences build the concepts required for subsequent intelligent behavior. He made some progress towards that end and pioneered the way for the vast armies of neuroscientists presently engaged in the task. What he succeeded in doing was to engineer a change in the philosophical outlook of behavioral scientists. Traditionally psychologists were dualists, believing that mental processes take place in an incorporeal mind; a "black box" whose workings could only be inferred by introspection or by observations of the

behavior it produced. Hebb offered a more direct window to the mind, investigation of its neural machinery. His monograph held the promise of a genuinely monist psychology.

5. Why Hebb remains important today

It is difficult to read a paper in the neurobiology of learning and memory today that does not mention its debt to Hebb. To cite but one example, Döbrössy and Dunnett (2001, p. 871) state that “The mechanisms by which environment and experience can influence brain structure and function first received direct experimental attention after Hebb’s observations and theoretical proposals that learning is represented in the brain by synaptic changes in response to repeated concurrent activation.” The reviews of Hebb’s ideas by Fentress (1999) and Milner (1999) show how Hebb has influenced research in many areas of psychology and neuroscience and how our findings have now gone beyond Hebb’s wildest speculations. Hebb’s cell assembly theory is becoming more important for guiding research (Sakurai, 1999) while the Hebb synapse remains a central component of molecular biological (Tsien, 2000) and computational (Klemm et al, 2000) models of learning. As stated by Ray Klein (1999, p. 3), “In the scientific literature, references to Hebb, the Hebbian cell assembly, the Hebb synapse and the Hebb rule increase each year. These forceful ideas of 1949 are now applied in engineering, robotics, and computer science as well as neurophysiology, neuroscience and psychology—a tribute to Hebb’s scientific acumen, foresight and courage to put forth a foundational neuropsychological theory of the organization of behaviour.”

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Preface

In this book I have tried to bring together a number of different lines of research, in a general theory of behavior that attempts to bridge the gap between neurophysiology and psychology, as well as that between laboratory psychology and the problems of the clinic.

The book is written, consequently, with the hope that it will be intelligible to the clinician and the physiologist as well as to psychologists. The development of psychological theory can be a cooperative affair, and has often been so in the past: witness Helmholtz, Jackson, Pavlov, Freud. The clinician and the physiologist frequently have direct access to data of first importance for psychology, sometimes without recognizing the fact. Though I have done my best, it may be chimerical to hope that my discussion is extensive and clear enough to stand on its own feet, for the nonpsychological reader. The reader who needs it will find more of the details of psychological theory in Morgan (1943) on psychophysiological psychology, Hilgard and Marquis (1940) on the theory of learning, Woodworth (1938) on “experimental” (normal human adult) psychology, and Moss (1942) or Maier and Schneirla (1935) on animal psychology. Of these, Morgan is most directly relevant, and in several places I have assumed a knowledge of fact to the extent provided by his text.

It is a pleasure to record my indebtedness to the colleagues who have read and improved the contents of this book. I owe much to students in a seminar at Harvard University in the summer of 1947, and in another at McGill University in the following winter. Part or all of the manuscript was read by Professor Harlow W. Ades, Professor Frank A. Beach, Dr. J. G. Beebe-Center, Professor R. B. MacLeod, Dr. Francis McNaughton, Dr. G. A. Miller, Dr.

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Finally I should like to express my gratitude to those who have helped with clerical work in the preparation of this book, in its various stages: particularly, Alice Sellers, Therese Mahoney, and Margaret Halliday. The McGill University Research Fund supported some of the experimental work that is briefly reported here, and also made it possible to secure additional clerical assistance.

D.O. HEBB

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Introduction

It might be argued that the task of the psychologist, the task of understanding behavior and reducing the vagaries of human thought to a mechanical process of cause and effect, is a more difficult one than that of any other scientist. Certainly the problem is enormously complex; and though it could also be argued that the progress made by psychology in the century following the death of James Mill, with his crude theory of association, is an achievement scarcely less than that of the physical sciences in the same period, it is nevertheless true that psychological theory is still in its infancy. There is a long way to go before we can speak of understanding the principles of behavior to the degree that we understand the principles of chemical reaction.

In an undertaking of such difficulty, the psychologist presumably must seek help wherever he can find it. There have been an increasing number of attempts to develop new mathematical methods of analysis. With these, in general, I do not attempt to deal. The method of factor analysis developed by Spearman (1927) and greatly elaborated by Thurstone (1935) is well established as a powerful tool for handling certain kinds of data, though the range of its use has been limited by dependence on tests that can be conveniently given to large groups of subjects. Another method is the application of mathematics more directly to the interaction of populations of neurons, by Rashevsky, Pitts, Householder, Landahl, McCulloch, and others.* Bishop (1946) has discussed the work from the point of view of neurophysiology, and his remarks are fully concurred with here. The preliminary

* Two papers by Culbertson (*Bull Math. Biophys.*, 1948, 10, 31–40 and 97–102), and Bishop's review article, list some of the more important of the actual titles in this field.

studies made with this method so far have been obliged to simplify the psychological problem almost out of existence. This is not a criticism, since the attempt is to develop methods that can later be extended to deal with more complex data; but as matters stand at present one must wait for further results before being sure that the attempt will succeed. Undoubtedly there is great potential value in such work, and if the right set of initial assumptions can be found it will presumably become, like factor analysis, a powerful ally of other methods of study.

However, psychology has an intimate relation with the other biological sciences, and may also look for help there. There is a considerable overlap between the problems of psychology and those of neurophysiology, hence the possibility (or necessity) of reciprocal assistance. The first object of this book is to present a theory of behavior for the consideration of psychologists; but another is to seek a common ground with the anatomist, physiologist, and neurologist, to show them how psychological theory relates to their problems and at the same time to make it more possible for them to contribute to that theory.

Psychology is no more static than any other science. Physiologists and clinicians who wish to get a theoretical orientation cannot depend only on the writings of Pavlov or Freud. These were great men, and they have contributed greatly to psychological thought. But their contribution was rather in formulating and developing problems than in providing final answers. Pavlov himself seems to have thought of his theory of conditioned reflexes as something in continual need of revision, and experimental results have continued to make revisions necessary: the theory, that is, is still developing. Again, if one were to regard Freud's theory as needing change only in its details, the main value of his work would be stultified. Theorizing at this stage is like skating on thin ice—keep moving, or drown. Ego, Id, and Superego are conceptions that help one to see and state important facts of behavior, but they are also dangerously easy to treat as ghostly realities: as anthropomorphic agents that *want* this or *disapprove* of that, *overcoming* one another by force or guile, and *punishing* or *being punished*. Freud has left us the task of developing these provisional formulations of his to the point where such a danger no longer

exists. When theory becomes static it is apt to become dogma; and psychological theory has the further danger, as long as so many of its problems are unresolved, of inviting a relapse into the vitalism and indeterminism of traditional thought.

It is only too easy, no matter what formal theory of behavior one espouses, to entertain a concealed mysticism in one's thinking about that large segment of behavior which theory does not handle adequately. To deal with behavior at present, one must oversimplify. The risk, on the one hand, is of forgetting that one has oversimplified the problem; one may forget or even deny those inconvenient facts that one's theory does not subsume. On the other hand is the risk of accepting the weak-kneed discouragement of the vitalist, of being content to show that existing theories are imperfect without seeking to improve them. We can take for granted that any theory of behavior at present must be inadequate and incomplete. But it is never enough to say, because *we* have not yet found out how to reduce behavior to the control of the brain, that no one in the future will be able to do so.

Modern psychology takes completely for granted that behavior and neural function are perfectly correlated, that one is completely caused by the other. There is no separate soul or life-force to stick a finger into the brain now and then and make neural cells do what they would not otherwise. Actually, of course, this is a working assumption only—as long as there are unexplained aspects of behavior. It is quite conceivable that some day the assumption will have to be rejected. But it is important also to see that we have not reached that day yet: the working assumption is a necessary one, and there is no real evidence opposed to it. Our failure to solve a problem so far does not make it insoluble. One cannot logically be a determinist in physics and chemistry and biology, and a mystic in psychology.

All one can know about another's feelings and awarenesses is an inference from what he *does*—from his muscular contractions and glandular secretions. These observable events are determined by electrical and chemical events in nerve cells. If one is to be consistent, there is no room here for a mysterious agent that is defined as not physical and yet has physical effects (especially since many of the entities of physics are known only through their

effects). "Mind" can only be regarded, for scientific purposes, as the activity of the brain, and this should be mystery enough for anyone: besides the appalling number of cells (some nine billion, according to Herrick) and even more appalling number of possible connections between them, the matter out of which cells are made is being itself reduced by the physicist to something quite unlike the inert stick or stone with which mind is traditionally contrasted. After all, it is that contrast that is at the bottom of the vitalist's objection to a mechanistic biology, and the contrast has lost its force (Herrick, 1929). The mystic might well concentrate on the electron and let behavior alone. A philosophical parallelism or idealism, whatever one may think of such conceptions on other grounds, is quite consistent with the scientific method, but interactionism seems not to be.

Psychologist and neurophysiologist thus chart the same bay—working perhaps from opposite shores, sometimes overlapping and duplicating one another, but using some of the same fixed points and continually with the opportunity of contributing to each other's results. The problem of understanding behavior is the problem of understanding the total action of the nervous system, and *vice versa*. This has not always been a welcome proposition, either to psychologist or to physiologist.

A vigorous movement has appeared both in psychology and psychiatry to be rid of "physiologizing," that is, to stop using physiological hypotheses. This point of view has been clearly and effectively put by Skinner (1938), and it does not by any means represent a relapse into vitalism. The argument is related to modern positivism, emphasizes a method of correlating observable stimuli with observable response, and, recognizing that "explanation" is ultimately a statement of relationships between observed phenomena, proposes to go to the heart of the matter and have psychology confine itself to such statements *now*. This point of view has been criticized by Pratt (1939) and Köhler (1940). The present book is written in profound disagreement with such a program for psychology. Disagreement is on the grounds that this arises from a misconception of the scientific method as it operates in the earlier stages. Those apparently naïve features of older scientific thought may have had more to do with hitting

on fertile assumptions and hypotheses than seems necessary in retrospect. The anti-physiological position, thus, in urging that psychology proceed now as it may be able to proceed when it is more highly developed, seems to be in short a counsel of perfection, disregarding the limitations of the human intellect. However, it is logically defensible and may yet show by its fertility of results that it is indeed the proper approach to achieving prediction and control of behavior.

If some psychologists jib at the physiologist for a bedfellow, many physiologists agree with them heartily. One must sympathize with those who want nothing of the psychologist's hairsplitting or the indefiniteness of psychological theory. There is much more certainty in the study of the electrical activity of a well-defined tract in the brain. The only question is whether a physiology of the human brain as a whole can be achieved by such studies alone. One can discover the properties of its various parts more or less in isolation; but it is a truism by now that the part may have properties that are not evident in isolation, and these are to be discovered only by study of the whole intact brain. The method then calls for learning as much as one can about what the parts of the brain do (primarily the physiologist's field), and relating behavior as far as possible to this knowledge (primarily for the psychologist); then seeing what further information is to be had about how the total brain works, from the discrepancy between (1) actual behavior and (2) the behavior that would be predicted from adding up what is known about the action of the various parts.

This does not make the psychologist a physiologist, for precisely the same reason that the physiologist need not become a cytologist or biochemist, though he is intimately concerned with the information that cytology and biochemistry provide. The difficulties of finding order in behavior are great enough to require all one's attention, and the psychologist is interested in physiology to the extent that it contributes to his own task.

The great argument of the positivists who object to "physiologizing" is that physiology has not helped psychological theory. But, even if this is true (there is some basis for denying it), one has to add the words *so far*. There has been a great access of knowledge in neurophysiology since the twenties. The work

of Berger, Dusser de Barenne, and Lorente de Nó (as examples) has a profound effect on the physiological conceptions utilized by psychology, and psychology has not yet assimilated these results fully.

The central problem with which we must find a way to deal can be put in two different ways. Psychologically, it is the problem of thought: some sort of process that is not fully controlled by environmental stimulation and yet cooperates closely with that stimulation. From another point of view, physiologically, the problem is that of the transmission of excitation from sensory to motor cortex. This statement may not be as much oversimplified as it seems, especially when one recognizes that the "transmission" may be a very complex process indeed, with a considerable time lag between sensory stimulation and the final motor response. The failure of psychology to handle thought adequately (or the failure of neurophysiology to tell us how to conceive of cortical transmission) has been the essential weakness of modern psychological theory and the reason for persistent difficulties in dealing with a wide range of experimental and clinical data, as the following chapters will try to show, from the data of perception and learning to those of hunger, sleep, and neurosis.

In mammals even as low as the rat it has turned out to be impossible to describe behavior as an interaction directly between sensory and motor processes. Something like *thinking*, that is, intervenes. "Thought" undoubtedly has the connotation of a human degree of complexity in cerebral function and may mean too much to be applied to lower animals. But even in the rat there is evidence that behavior is not completely controlled by immediate sensory events: there are central processes operating also.

What is the nature of such relatively autonomous activities in the cerebrum? Not even a tentative answer is available. We know a good deal about the afferent pathways to the cortex, about the efferent pathways from it, and about many structures linking the two. But the links are complex, and we know practically nothing about what goes on between the arrival of an excitation at a sensory projection area and its later departure from the motor area of the cortex. Psychology has had to find, in hypothesis,

a way of bridging this gap in its physiological foundation. In general the bridge can be described as some comparatively simple formula of cortical transmission.* The particular formula chosen mainly determines the nature of the psychological theory that results, and the need of choosing is the major source of theoretical schism.

Two kinds of formula have been used, leading at two extremes to (1) switchboard theory, and sensori-motor connections; and (2) field theory. (Either of these terms may be regarded as opprobrium; they are not so used here.) (1) In the first type of theory, at one extreme, cells in the sensory system acquire connections with cells in the motor system; the function of the cortex is that of a telephone exchange. Connections rigidly determine what animal or human being does, and their acquisition constitutes learning. Current forms of the theory tend to be vaguer than formerly, because of effective criticism of the theory in its earlier and simpler forms, but the fundamental idea is still maintained. (2) Theory at the opposite extreme denies that learning depends on connections at all, and attempts to utilize instead the field conception that physics has found so useful. The cortex is regarded as made up of so many cells that it can be treated as a statistically homogeneous medium. The sensory control of motor centers depends, accordingly, on the distribution of the sensory excitation and on ratios of excitation, not on locus or the action of any specific cells.

Despite their differences, however, both theoretical approaches seem to imply a prompt transmission of sensory excitation to the motor side, if only by failing to specify that this is not so. No one, at any rate, has made any serious attempt to elaborate ideas of a central neural mechanism to account for the delay, between stimulation and response, that seems so characteristic of thought. There have indeed been neural theories of "motor" thought, but they amount essentially to a continual interplay of proprioception and minimal muscular action, and do not provide for any prolonged sequence of intracerebral events as such.

* The simplicity possibly accounts for the opinion expressed by an anatomist who claimed that psychologists think of the brain as having all the finer structure of a bowlful of porridge.

But the recalcitrant data of animal behavior have been drawing attention more and more insistently to the need of some better account of central processes. This is what Morgan (1943) has recognized in saying that "mental" variables, repeatedly thrown out because there was no place for them in a stimulus-response psychology, repeatedly find their way back in again in one form or another. The image has been a forbidden notion for twenty years, particularly in animal psychology; but the fiend was hardly exorcised before "expectancy" had appeared instead. What is the neural basis of expectancy, or of attention, or interest? Older theory could use these words freely, for it made no serious attempt to avoid an interactionist philosophy. In modern psychology such terms are an embarrassment; they cannot be escaped if one is to give a full account of behavior, but they still have the smell of animism: and must have, until a theory of thought is developed to show how "expectancy" or the like can be a physiologically intelligible process.

In the chapters that follow this introduction I have tried to lay a foundation for such a theory. It is, on the one hand and from the physiologist's point of view, quite speculative. On the other hand, it achieves some synthesis of psychological knowledge, and it attempts to hold as strictly as possible to the psychological evidence in those long stretches where the guidance of anatomy and physiology is lacking. The desideratum is a conceptual tool for dealing with expectancy, attention, and so on, and with a temporally organized intracerebral process. But this would have little value if it did not also comprise the main facts of perception, and of learning. To achieve something of the kind, the limitations of a schema are accepted with the purpose of developing certain conceptions of neural action. This is attempted in Chapters 4 and 5; Chapters 1 to 3 try to clear the ground for this undertaking. From Chapter 6 onward the conceptions derived from schematizing are applied to the problems of learning, volition, emotion, hunger, and so on. (In general, the reader may regard Chapters 1 to 5 as mainly preparatory, unless he is particularly interested in the neurological details, or in the treatment of perception; to get the gist of the theory that is presented here one should read the two following paragraphs, and turn directly to Chapter 6.) In outline, the conceptual structure is as follows:

Any frequently repeated, particular stimulation will lead to the slow development of a "cell-assembly," a diffuse structure comprising cells in the cortex and diencephalon (and also, perhaps, in the basal ganglia of the cerebrum), capable of acting briefly as a closed system, delivering facilitation to other such systems and usually having a specific motor facilitation. A series of such events constitutes a "phase sequence"—the thought process. Each assembly action may be aroused by a preceding assembly, by a sensory event, or—normally—by both. The central facilitation from one of these activities on the next is the prototype of "attention." The theory proposes that in this central facilitation, and its varied relationship to sensory processes, lies the answer to an issue that is made inescapable by Humphrey's (1940) penetrating review of the problem of the direction of thought.

The kind of cortical organization discussed in the preceding paragraph is what is regarded as essential to adult waking behavior. It is proposed also that there is an alternate, "intrinsic" organization, occurring in sleep and in infancy, which consists of hypersynchrony in the firing of cortical cells. But besides these two forms of cortical organization there may be disorganization. It is assumed that the assembly depends completely on a very delicate timing which might be disturbed by metabolic changes as well as by sensory events that do not accord with the pre-existent central process. When this is transient, it is called emotional disturbance; when chronic, neurosis or psychosis.

The theory is evidently a form of connectionism, one of the switchboard variety, though it does not deal in direct connections between afferent and efferent pathways: not an "S-R" psychology, if R means a *muscular* response. The connections serve rather to establish autonomous central activities, which then are the basis of further learning. In accordance with modern physiological ideas, the theory also utilizes local field processes and gradients, following the lead particularly of Marshall and Talbot (1942). It does not, further, make any single nerve cell or pathway essential to any habit or perception. Modern physiology has presented psychology with new opportunities for the synthesis of divergent theories and previously unrelated data, and it is my intent to take such advantage of these opportunities as I can.

1.

The Problem and the Line of Attack

This book presents a theory of behavior that is based as far as possible on the physiology of the nervous system, and makes a sedulous attempt to find some community of neurological and psychological conceptions.

The purpose of this first chapter is to define the major difficulties in the way of such an undertaking and to show how these difficulties determine the line it must take. The great need, as I have tried to make clear in the Introduction, is for a physiological theory of thought (or some equivalent term that can be applied to animals as well as to man).

Let us look first at the specific problem from which the present speculations began, to see why the discussion is oriented as it is.

The problem lay in certain puzzling effects of operation on the human brain. The effect of a clearcut removal of cortex outside the speech area is often astonishingly small; at times no effect whatever can be found (Hebb, 1942*a*, 1945*b*). It is possible that there is always a loss of intelligence in aphasia, when the “speech area” is seriously damaged, but this does not, of course, explain why damage elsewhere should have no effect. It would be unreasonable to suppose that most of the cortex has nothing to do with intelligence, and there are in fact definite indications that this is not true. Intelligence must be affected by any large brain injury—yet sometimes it seems not to be.

A final touch is added to the puzzle when we find that it is tests of the Binet type that least show the effect of injuries outside the speech area. The Binet is the measure of intelligence that is most sensitive and accurate with normal subjects. How can it be possible for a man to have an IQ of 160 or higher, after a prefrontal lobe

has been removed (Hebb, 1939), or for a woman (described by Rowe, 1937) to have an IQ of 115, a better score than two-thirds of the normal population could make, after losing the entire right half of the cortex?

Those two are perhaps the most striking cases, but high scores after brain operation have been reported by nearly everyone who has used standard tests as a method of study, and they have to be explained. Most investigators have preferred to forget them, and to search only for intellectual defects. The defects certainly must exist, and it is important to find them; but it is just as important to learn why Binet-type tests, the most valid and sensitive indices of normal ability, should often show no effect of injury to the brain.

The only explanation that has been proposed, and perhaps the only feasible one, has to do with perceptual learning and with concepts as distinct from conditioned responses or motor learning (Hebb, 1942*a*). The explanation, roughly, is this. The level of intelligence-test performance is a function of the concepts a patient has already developed. Once developed, a concept is retained, despite brain damage that, if it had occurred earlier, would have prevented the development. The patient with brain injury at maturity may continue to think and solve problems normally (in familiar fields), although his intelligence would have been far from normal if a similar injury had happened at birth. The explanation meets the clinical facts and, moreover, is supported by the way in which some intellectual capacities are retained in old age when others are disappearing.

But now we come to the crux of the matter. As an explanation, this is only a good start. What is a concept, physiologically and *à propos* of the loss of neural cells? Though it has been tried before now, an explanation cannot be half neural anatomy and half consciousness. They are different levels in a logical hierarchy. A problem recognized at one level can be solved by recourse to a more fundamental set of conceptions, but an hypothesis cannot well comprise two levels, take in two universes of discourse at the same time. We want to explain certain clinical facts. To really do so, we must find an anatomical and physiological understanding of what is known psychologically as a concept; and we must be able to deal with its relation to perception and to learning.

And with *that*, we land right in the middle of the generalized problem of explaining mammalian behavior. What is a concept, if it is not a conditioned response? What is perceptual learning? And so on. Before such questions can be answered, psychological theory must have a new base of operations. As an illustration, a particular problem has been cited; but the difficulty is really general, as the following section will try to show, and there is no phase of psychological theory in which the same central weakness does not appear.

It has already been suggested that the essential need is to find out how to handle thought, and related processes, more adequately. The difficulty in doing so goes back to fundamental assumptions. If this discussion is to rest on solid ground, it must start with certain ideas with which every psychologist is concerned but which have been confused, vague, or ill defined.

This first chapter, accordingly, will dot some *i*'s and cross some *t*'s, so that we can get down to business in what follows. There are two fundamental assumptions to be dealt with. One of them must be rejected, one accepted. The rest of this chapter will show how they determine a point of attack for the revision of theory.

REJECTING THE ASSUMPTION OF A COMPLETE SENSORY CONTROL

The first to be discussed is what can be called the assumption of a sensory dominance of behavior. It is the idea that behavior is a series of *reactions* (instead of actions), each of which is determined by the immediately preceding events in the sensory systems. The idea is not altogether consistent with recognizing the existence of set, attitude, or attention; and an implicit inconsistency of this sort is at the root of the current confusion in psychological theory.

It may be noted in passing that the assumption of a sensory dominance of behavior is not the property of any particular theory. Theories differ as to how a sensory event has its effect, but not as to its all-important role. *Gestalt* psychology avoids words like "stimulus" and "sensation" but as a system is not less preoccupied with stimulus configurations than other systems are with their stimuli (p. 23).

Now for the source of confusion.

In the simplest terms, "attention" refers to a selectivity of response. Man or animal is continuously responding to some events in the environment, and not to others that could be responded to (or "noticed") just as well. When an experimental result makes it necessary to refer to "set" or "attention," the reference means, precisely, that the activity that controls the form, speed, strength, or duration of response is not the immediately preceding excitation of receptor cells alone. The fact that a response is not so controlled may be hard to explain, theoretically; but it is not mystical, and "attention" is not necessarily anthropomorphic, or animistic, or undefinable.

Now the tradition in psychology has long been a search for the property of the stimulus which by itself determines the ensuing response, at any given stage of learning. This approach seems partly a consequence of psychology's persistent fight against animism* and deserves respect for that reason; but it is no longer satisfactory as theory. Almost without exception psychologists have recognized the existence of the selective central factor that reinforces now one response, now another. The problem is to carry out to its logical conclusion an incomplete line of thought that starts out preoccupied with stimulus or stimulus configuration as the source and control of action, eventually runs into the facts of attention and so on, and then simply agrees that attention is an important fact, without recognizing that this is inconsistent with one's earlier assumptions. To complete this process, we must go back and make a change in the basis of the theory.

There are three points here: one is that psychologists have generally recognized the existence of attention or the like; another

* And partly, as we shall see, the product of antiquated physiological conceptions. Fighting animism meant that psychological phenomena had to be reduced to a pattern of cause and effect. The only means at hand was a physiology of the nervous system in which a knowledge of sense organs and peripheral nerve was the main content. As a result, it is still easy to feel that psychology becomes part of a larger demonology with any retreat from the stimulus-response formula. As for "insight," "purpose," "attention"—any one of these may still be an invocation of the devil, to the occasional psychologist. However, this attitude should not be too much made fun of. It cannot be emphasized too strongly that there is continual danger of slipping momentarily into animistic thinking; and consistent use of the S-R formula does at least avoid that danger effectively. It behooves those of us who wish to use other terms to see that they are clearly defined.

that they have done so reluctantly and sparingly, and have never recognized the fact in setting up theories. The third point is obvious enough, that we need to find some way of dealing with the facts consistently. Since everyone knows that attention and set exist, we had better get the skeleton out of the closet and see what can be done with it.

The first two of these points have been pretty clearly established by Gibson (1941). His review needs some clarification in one respect, since he declined to attempt any definition of "set" or any other of the long list of terms with a similar meaning that he gathered together, although he evidently recognized, in classifying them so, that they have something in common. This common meaning has already been defined. When one considers the problem in the light of the implicit assumption of a sensory dominance of behavior it becomes clear at once that the notions of set, attention, attitude, expectancy, hypothesis, intention, vector, need, perseveration, and preoccupation (Gibson, 1941, pp. 781–782) have a common element, and one only. That element is the recognition that responses are determined by something else besides the immediately preceding sensory stimulation. It does not deny the importance of the immediate stimulus; it does deny that sensory stimulation is everything in behavior.

All such terms, then, are a reference to the "central process which seems relatively independent of afferent stimuli," defined by Hilgard and Marquis (1940), which I shall call here the *autonomous central process*. Gibson's review shows in effect that the process is ubiquitous, that it crops up in every sort of psychological investigation—and that almost everyone has recognized its existence, in one form or another. To Gibson's list can be added Pavlov's (1928) and Hull's (1943) stimulus trace as lasting cerebral state, set up it is true by a specific stimulus but not transmitted and dissipated at once; Beach's (1942) central excitatory mechanism; Morgan's (1943) central motive state; and Kleitman's (1939) "interest"—a factor in wakefulness. All these things have the same property of an activity that has a selective effect on behavior without being part of the present afferent excitation.

Everyone has had such ideas about the control of behavior; and yet, as Gibson noted, "The meaning [of the term "set"] is

felt to be unsatisfactory, and the concept is employed reluctantly and only because the facts make it absolutely unavoidable." The reluctance is partly no doubt because of a feeling that the concept is animistic, in some obscure way. But why animism, if the facts of behavior make it unavoidable? The trouble really seems to have been in finding how to make an essential idea intelligible.

Hilgard and Marquis' central process, and Beach's central excitatory mechanism, are hypothetical entities, but they certainly have no flavor of animism about them. "Attention" and "set" are now seen to fall in the same class: it may well be that their connotations are misleading and that we shall have to look for new terms, but the idea itself is respectable, and such language need no longer risk starting a witchhunt.

THE NEUROLOGICAL PROBLEM OF ATTENTION

There is a further hazard on the course. This is the apparent lack of a theoretical rationale for the autonomous central process. Actually, modern neurophysiology has already removed this difficulty.

Here again the situation can only be understood historically. A main function of the neural cell is of course to transmit excitations, and earlier ideas of anatomy and physiology made the central nervous system appear, in principle, a collection of routes, some longer, some shorter, leading without reversal from receptors to effectors—a mass of conductors that lies inactive until a sense organ is excited, and then conducts the excitation promptly to some muscle or gland. We know now that this is not so, but the older idea still has a profound effect on psychological thought—demonstrated, for example, in the assumption of sensory dominance, discussed above.

The lack of a rationale for nonsensory influences on behavior that seemed to exist in 1920 certainly exists no more. Psychologists have long had to recognize (since the days of the Würzburg school at least) the existence of a determining tendency, whether physiology made it comprehensible or not. Modern electrophysiology has more than caught up with psychology and now provides abundant evidence to support the same idea. When the detailed evidence of neurophysiology and histology are

considered, the conclusion becomes inevitable that the nonsensory factor in cerebral action must be more consistently present and of more dominating importance than reluctant psychological theory has ever recognized. Instead of a joker to occasionally confuse the student of behavior, nonsensory activities appear in every fall of the cards and must make up a large share of the deck. Neurophysiologically, it may even become a problem to account for *any* consistent effect of a specific stimulus (pp. 121, 126).

Electrophysiology of the central nervous system indicates in brief that the brain is continuously active, in all its parts, and an afferent excitation must be superimposed on an already existent excitation. It is therefore impossible that the consequence of a sensory event should often be uninfluenced by the pre-existent activity. If we recognize in that activity the psychologically known factor of set and the like (as Denny-Brown, 1932, suggested some time ago), the problem for psychology is no longer to account for the existence of set but to find out how it acts and above all to learn how it has the property of a consistent, selective action instead of producing the random-error distribution postulated by Hull (1943) in his "oscillation principle."

So there really is a rational basis for postulating a central neural factor that modifies the action of a stimulus. The theoretical problem now is to discover the rules by which it operates. At first glance this is a problem for the neurophysiologist only. But look closer; much of the evidence, from which these rules must be worked out, is psychological, or behavioral. The problem is after all the problem of attention, and seen best in the activity of the whole animal. It is in the highest degree unlikely that it can be solved either from the physiological evidence alone or from the behavioral evidence alone. What we need, evidently, is some synthesis of both kinds of datum. The psychological data have been reviewed briefly (for a fuller account, see Hilgard and Marquis, 1940; Gibson, 1941; Morgan, 1943—under such headings as *attitude*, *expectancy*, and *set*). Let us turn now, again briefly, to the anatomical and physiological evidence that has made a drastic change in the theoretical problems of behavior.

For our purposes, the physiological evidence can be treated under two heads, as bearing on (1) the existence and properties

of a continuous cerebral activity, and (2) the nature of synaptic transmission in the central nervous system.

1. The studies reviewed by Jasper (1937), beginning with Berger's work and its confirmation by Adrian, have shown with practical certainty that the central nervous system is continuously active, in all its parts, whether exposed to afferent stimulation or not. It appears that the EEG, or electroencephalogram, must be at least in part a summation of action potentials, an index of actual cellular firing. There are changes of cell potential without active transmission of impulses (Gibbs, 1945), and it is quite possible that these are the source of much of the EEG. However, there is a considerable body of evidence to show that neural tissue is persistently active; and presumably the EEG includes a record of that activity. *It is taken here as a working assumption that the EEG is correlated with neural firing*—that large potentials indicate a local synchrony of firing, even though other factors contribute to the size of the potential. The psychological usefulness of this assumption will perhaps become evident in the following chapters.

The evidence concerning persistent activity consists first of a number of studies showing directly that the neural cell fires spontaneously, *i.e.*, under no stimulation except that of the nutrient fluids bathing it (Fessard, cited by Jasper, 1937; Prosser, 1934; Lehmann, 1937*a*, 1937*b*; Bronk, 1939; Dubner and Gerard, 1939; Libet and Gerard, 1939). Secondly, a very important paper for the theory of behavior, by Weiss (1941*a*), reports spontaneous, almost incessant motor activity in a transplanted amphibian limb innervated from a pool without afferent fibers. The activity of the limb was not normally coordinated, which agrees with the general idea that sensory activity is essential to the regulation of central neural firing but not essential to initiating it. Finally, the work of Adrian and his collaborators: Adrian and Matthews (1934) rejected the earlier opinion of Adrian (1931) and Adrian and Buytendijk (1931), and concluded definitely that the EEG is the summation of single sharp potentials, the result of actual cellular firing. They concluded further that the activity is not necessarily maintained by sensory activity. Elsewhere Adrian (1934, p. 1126) has stated his general conclusions: "There are

cell mechanisms in the brain which are set so that a periodic discharge is bound to take place. The moment at which it occurs can be greatly altered by afferent influences, but it cannot be postponed indefinitely."

Several comments may be made on these conclusions. Spontaneity of firing by central neural cells is not philosophic indeterminacy, as some writers have thought; the "spontaneity" means only that synaptic stimulation is not the sole cause of firing. As Adrian pointed out, also, the cell that is capable of firing spontaneously is also open to synaptic control. If it is left undisturbed the processes of metabolism will lead in the end to firing, but obviously an afferent stimulation could always catch the cell in the period between the absolute refractory state and the time of spontaneous discharge. With a frequent arrival of impulses at the synapse, therefore, the cell can never reach the point of spontaneous action but must remain under afferent control.

A very interesting relationship exists between sensory activity, EEG, and coordinated, adaptive behavior (or "conscious state"). There is, in the first place, Adrian and Matthews' observation that sensory processes, instead of supporting synchronous, rhythmic firing and large potentials in the EEG, have the opposite effect. They introduce irregularity and flattening of the electrical record. In the second place, large potentials, or "hypersynchrony," negate or may negate normal function (Jasper, 1941). That is, sensory activity breaks up hypersynchrony and makes for normal, coordinated, adaptive activity. Bartley and Bishop (1933) and Adrian and Matthews have even proposed that anesthetics produce "unconsciousness" by suppressing sensory activity rather than acting directly on the cerebrum itself. It does seem clear from the facts discussed, from the large potentials observed in sleep, and from the hypnotic effect of minimizing the normal variation of sensory activity, that the sensory input to the brain has a constantly necessary function, for adaptive behavior. We shall return to this point later (p. 145). For the present, it has been advisable to point out that recognizing the existence of spontaneous neural activity is neither indeterminacy nor a denial of the importance of sensory processes in normal behavior.

2. The nature of synaptic transmission in the central nervous system is also of fundamental importance for the theory of behavior. There are two radical modifications of earlier ideas: transmission is not simply linear but apparently always involves some closed or recurrent circuits; and a single impulse cannot ordinarily cross a synapse—two or more must act simultaneously, and two or more afferent fibers must therefore be active in order to excite a third to which they lead.

The concepts of neural action chiefly developed by Lorente de Nó (1938a, 1938b, 1939, 1943) are well enough known by now to need no elaborate review. It is necessary, however, to point out that they have revolutionary implications for psychological theory.

In particular, the psychological criticism advanced by Koffka (1924), Lashley (1929b, 1930), and Köhler (1929), against the theory of neural connections as the basis of learning, applies only to the older theory of *linear*, sensori-motor connections, in which a single cell was supposed to be always capable of exciting a second cell with which it synapsed. The criticism is effective against the idea that synaptic resistances are all that determines the direction of transmission at the synapse. A radical change in the whole problem, however, has been made by considerations of timing in transmission (Gasser, 1937) and by the conception of “optional transmission” (Lorente de Nó, 1939).

In a single system, and with a constant set of connections between neurons in the system, the direction in which an entering excitation will be conducted may be completely dependent on the timing of other excitations. Connections are necessary but may not be decisive in themselves; in a complex system, especially, time factors must always influence the direction of conduction. The older ideas of neural transmission gave synaptic connections too much rigidity, as a determinant of behavior. The rigidity has now disappeared; the idea of connections can again be useful in psychological theory, and the question of “synaptic resistances” is completely reopened.

Let me now summarize what has been said, in this and the preceding section, about attention, set, attitude, and so on. (1) All psychologists have recognized some such factor in behavior.

It undoubtedly exists. (2) Recognizing it is really a denial that behavior is only a series of responses to environmental stimulation. One important meaning of "attention" or the like is the reference to a partly autonomous, or nonsensory, cerebral activity: the "autonomous central process." (3) The problem for psychology then is to find conceptions for dealing with such complexities of central neural action: conceptions that will be valid physiologically and at the same time "molar" enough to be useful in the analysis of behavior. (4) Psychology is still profoundly influenced by the very "molecular" conception of linear transmission through a sequence of single cells. The conception is no longer valid physiologically, just as it has long been without psychological usefulness. The attack on neural connections as an explanation of behavior was really an attack on this particular conception of the way connections operate; modern neuroanatomy and electrophysiology have changed the question completely, and the significance of synaptic connections must be examined all over again.

Our problem, then, is to find valid "molar" conceptions of neural action (conceptions, *i.e.*, that can be applied to largescale cortical organizations). Bishop (1946, p. 370) has made the point, in another context, that this is an essential problem for neurophysiology also. But psychologists can hardly sit around with hands folded, waiting for the physiologist to solve it. In its essence the problem is psychological and requires a knowledge of the psychological as well as the physiological evidence for its solution.

PERCEPTUAL GENERALIZATION AND THE ASSUMPTION OF A STRUCTURAL MEMORY TRACE

Now for a second fundamental assumption of psychological theory: this time, one that must, it seems, be accepted; but in accepting it we must also recognize the difficulties it entails and provide for them. These difficulties in fact determine the main features of the theory presented in this monograph.

The assumption we must accept is that the memory trace, the basis of learning, is in some way structural and static; and the

difficulties in the way of making the assumption are mainly in the facts of perceptual generalization that have been emphasized by *Gestalt* psychologists (Koffka, 1935; Köhler, 1929, 1940) and Lashley (1938*b*, 1942*a*). The problem raised by these writers is crucial and must be disposed of before we touch anything else.

Lashley has concluded that a learned discrimination is not based on the excitation of any particular neural cells. It is supposed to be determined solely by the pattern, or shape, of the sensory excitation. Köhler, also stressing the apparent fact that the pattern and not the locus of stimulation is the important thing, has developed a theory of electrical fields in the brain which control cerebral action. Like Lashley, he explicitly denies that the same cells need be excited to arouse the same perception.

This suggests that the mnemonic trace, the neural change that is induced by experience and constitutes "memory," is not a change of structure. Other facts, at the same time, are an even stronger argument that it *must* be structural. A structural trace, as we shall see in a moment, must be assumed; but when we do so we have to find some way of fitting in the facts of perception.

If it is really unimportant in what tissues a sensory excitation takes place, one finds it hard to understand how repeated sensations can reinforce one another, with the lasting effect we call learning or memory. It might be supposed that the mnemonic trace is a lasting pattern of reverberatory activity without fixed locus, like some cloud formations or an eddy in a millpond. But if so it seems that the multitudinous traces in the small confines of the cerebral cortex would interfere with one another, producing a much greater distortion of early memories by later ones than actually occurs.

Moreover, violent cortical storms can occur (as in *grand mal* epilepsy or cerebral concussion) without a detectable effect on earlier memories. That the trace should be purely "dynamic"—a pattern of activity not dependent on structural changes for its permanence—thus seems in the highest degree unlikely. No one has explicitly made such an assumption; yet how otherwise are the known properties of a learned discrimination to be accounted for, with its inevitable tendency to be generalized beyond what has already been experienced by the animal—its apparent independence of excitation in specific cells?

In addition to the facts of perceptual generalization, two other forms of evidence might make it difficult to postulate a structural trace as the basis of memory. One is from Lashley's (1929*a*) extirpation experiments, showing that the removal of blocks of the rat's cerebral cortex does not affect habits selectively. If one habit is affected, others are also. From this, Lashley has concluded that memory traces are not localized in the cerebral cortex, but himself has pointed out (Lashley, 1929*b*) another possible interpretation. His evidence is consistent with the idea that the trace is structural but diffuse, involving, that is, a large number of cells widely spaced in the cortex, physiologically but not anatomically unified. This is not, consequently, crucial evidence for or against the notion of structural traces in the cortex.

The other evidence that seemed once to prevent postulating a structural trace is found in the work of Wulf (cited by Koffka, 1935) and later investigators who have interpreted their studies of human memory for patterns to mean that the trace is spontaneously active, and does not lie dormant or merely deteriorate with the passage of time. Hanawalt (1937), however, effectively criticized the earlier evidence for this idea; and Hebb and Foord (1945), having obtained data inconsistent with Wulf's hypothesis, re-examined the later work that managed to avoid Hanawalt's criticism. They have shown that there is no evidence to even faintly support the idea of slow, spontaneous changes in the trace. This conception must be abandoned.

Thus the only barrier to assuming that a structural change in specific neural cells is the basis of memory lies in the generalization of the perception of patterns. Man sees a square as a square, whatever its size, and in almost any setting. A rat trained to look for food behind a horizontal rectangle will there- after choose almost *any* horizontal figure, such as an interrupted line or a pair of circles side by side (figure 1). Trained to choose a solid upright triangle and to avoid an inverted triangle, he will discriminate consistently between outlines of triangles; triangles with confusing figures added (such as circumscribing circles); and triangles of different size, which cannot thus excite the same retinal cells simultaneously (Lashley, 1938*b*). Rats reared in darkness, then

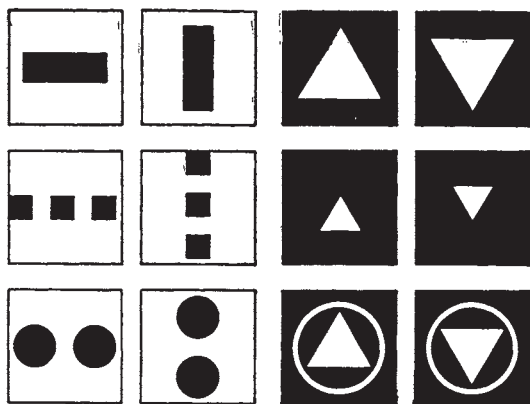


FIGURE 1. Illustrating generalization in perception. A rat is first *trained* with the two diagrams at the top, in the left half of the figure, until he always avoids the vertical bar and chooses the horizontal. He is then *tested* with the next pair of diagrams, the horizontally and vertically arranged squares; and with the next pair, circles side by side versus circles one above the other. Transfer of response occurs, the rat in each case preferring the horizontal diagram, showing that perception of horizontal and vertical is generalized beyond any particular pattern. Similarly, after training with the large plain triangles, top right, the animal is tested with smaller triangles and with circumscribed triangles. If the erect triangle, the one to the left, is positive in training, the rat will choose the left-hand diagram when presented with either of the two lower pairs, again demonstrating perceptual generalization.

trained in the same way, show the same perceptual generalizations (Hebb, 1937a).

These are concrete, undisputed facts of behavior. They have been interpreted as meaning that perception is independent of the locus of excitation; and this interpretation has been tacitly accepted as inescapable. The result is an awkward dilemma for theory, since, as we have seen, it is hard to reconcile an unlocalized afferent process with a structural (and hence localized) mnemonic trace.

Lashley's (1942a) hypothesis of interference patterns is the one explicit attempt to solve this difficulty and to deal adequately with both perception and learning. As such it deserves special mention

here, although we shall see that in other respects it faces great difficulties.

Other writers have had to choose one horn of the dilemma. Köhler (1940), for example, starts out with the facts of perceptual generalization, in his theory of cerebral fields of force, and then cannot deal with learning. He has no apparent way of avoiding a fatal difficulty about the nature of the trace, its locus and structure. This is another aspect of the difficulty for *Gestalt* theory raised by Boring (1933), who pointed out that at *some* point the perceptual process must act on specific cells in order to determine a specific response.

The theory elaborated by Hull (1943), on the other hand, is to be regarded as providing first of all for the stability of learning. It then has persistent difficulty with perception. The principle of "afferent neural interaction" appears to be a concession extorted by the facts of perceptual generalization. With this, there is some danger that the entire system may lose its meaning. The great value of Hull's theory is in showing how one may conceive of variable behavior as determined by constant causal relationships between stimulus, intervening variables, and response. This is brilliantly achieved, for an important segment of behavior. But then the postulate of afferent neural interaction adds that anything may happen when two sensory events occur at the same time—which of course they are always doing. Evidently no prediction is possible until the limits, and the determinants, of afferent neural interaction can be given in detail. This it seems demands that the neurological reference, already present in the theory, be made explicit, and detailed. For our present purposes, at any rate, Hull must be regarded as not yet having solved the problem of dealing with the perceptual process in a theory of learning, although it remains possible that his program (Hull, 1945) will do so in the future.

THE MODE OF ATTACK

How are we to provide for perceptual generalization *and* the stability of memory, in terms of what the neuron does and what happens at the synapse? We must suppose that the mnemonic

trace is a structural change; the difficulty, in supposing it, is a conflict with the idea that only the pattern, and not the locus of sensory stimulation, is important in perception; so let us begin by asking whether that idea is, after all, securely established.

That paragraph puts the theoretical approach of this monograph. I propose to go over the evidence on perception again and show that it is not what it seems. We do *not* know that pattern is everything, locus nothing. Next, an alternative set of ideas about perception will be developed. According to these ideas, perception does depend on exciting specific parts of the receptor surface; mnemonic trace and perceptual generalization need no longer be at odds with one another; and a physiological meaning of attention (or set, or expectancy) shows up.

Now let us see, in the following two chapters, what revisions can be made in current ideas about perception so that they can be extended into a more general theory. The reader will observe that the discussion deals almost entirely with *visual* perception. This is not because vision has any unique significance, but because it is in visual perception, with few exceptions, that the problem of patterning and form has been studied experimentally. It can be assumed that perception in other sense modes does not depend on fundamentally different principles.

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